

The Quest of the Hujinator

Part 1

Nom	TeachCS (A)	GiveBonus (B)	HasGlasses (C)	Beloved
Prof 1	1	1	1	Y
Prof 2	1	1	1	Y
Prof 3	1	0	1	N
Prof 4	1	0	0	N
Prof 5	0	1	1	Y
Prof 6	0	1	0	Y
Prof 7	0	0	0	N
Prof 8	0	0	0	N
Prof 9	1	1	1	Y
Prof 10	0	1	1	Y

1. Initial entropy:

$$H(S) = -(0.6 \times \log_2 0.6 + 0.4 \times \log_2 0.4) = 0.6 \times 0.74 + 0.4 \times 1.32 = 0.444 + 0.528 = 0.972$$

2. Information Gain:

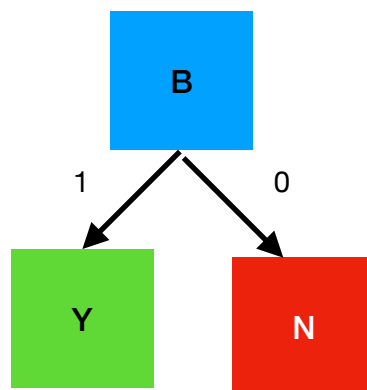
A. $Gain(S, A) = H(S) - 0.5(0.6 \log_2 0.6 + 0.4 \log_2 0.4) - 0.5(0.6 \log_2 0.6 + 0.4 \log_2 0.4) = 0$
No gain of information, splitting by A doesn't decrease the entropy

B. $Gain(S, B) = H(S) - 0.5(1 \log_2 1) - 0.5(1 \log_2 1) = 0.972$
Gain value is the same as entropy, by splitting by B the Beloved column is completely predictable,

C. $Gain(S, C) = H(S) - 0.6(\frac{5}{6} \log_2 \frac{5}{6} + \frac{1}{6} \log_2 \frac{1}{6}) - 0.4(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4}) = 0.972 - 0.714 = 0.2565$

C can contribute for entropy reduction.

3. Decision Tree: feature B will be used as root node, line all data was labeled correctly no further nodes are needed.



4. C will be chosen as the root node since the IG for that feature is non-zero and higher than IG of A.

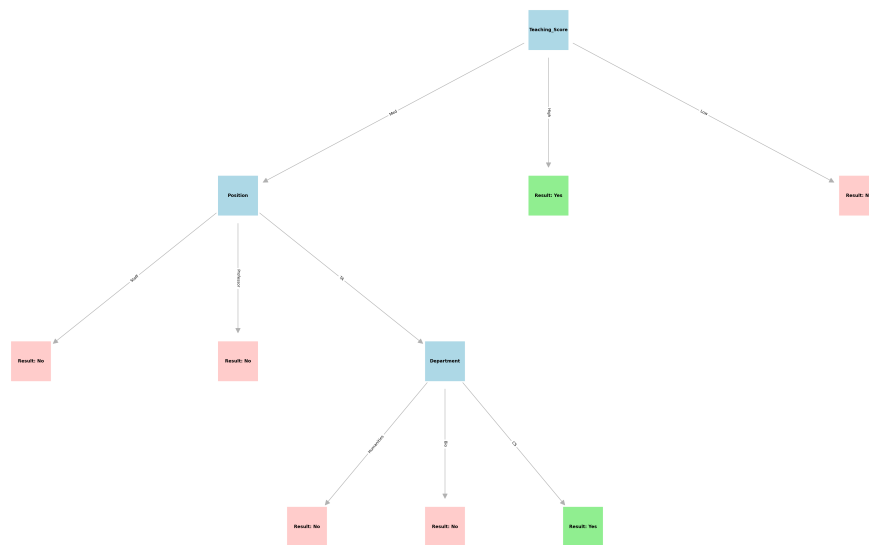
Part 2

Task A bug fix:

1. Line 18: missing minus before the sum of the entropy, bug effectively lower the IG value
2. Line 29: wrong weight, was constant 2, changed to the fraction of the current value in the loop
3. Line 32: wrong calculation, should be the positive diff between entropy and weighted entropy
4. Line 44: add a stop to max_depth parameter, not relevant if code get. To a different stop condition.
5. Line 52: setting threshold for minimum gain to split, can be adjusted
6. Line 85: pass max_depth=5, min_samples_split=10 to build_tree, fail without setting values.

Task B:

Minimal tree:



Hujinator decision tree

From the this tree we get that high teaching score is correlated with being beloved. And there are some beloved teachers with medium score and are also TAs in CS Department